55.2: Energy and other limiting factors control primary production in ecosystems

The theme of energy transfer underlies all biological interactions (see Concept 1.1). In most ecosystems, the amount of light energy converted to chemical energy—in the form of organic compounds—by autotrophs during a given time period is the ecosystem's **primary production**. In ecosystems where the primary producers are chemoautotrophs, the initial energy input is chemical, and the initial products are the organic compounds synthesized by the microorganisms.

Ecosystem Energy Budgets

In most ecosystems, primary producers use light energy to synthesize energy-rich organic molecules, and consumers acquire their organic fuels secondhand (or even third- or fourth-hand) through food webs (see Figure 54.17). Therefore, the total amount of photosynthetic production sets the "spending limit" for the entire ecosystem's energy budget.

The Global Energy Budget

Each day, Earth's atmosphere is bombarded by approximately 1022joules of solar radiation (1J=0.239 cal). This is enough energy to supply the demands of the entire human population for 19 years at 2013 energy consumption levels. The intensity of the solar energy striking Earth varies with latitude, with the tropics receiving the greatest input (see Figure 52.3). About 50% of incoming solar radiation is absorbed, scattered, or reflected by clouds and dust in the atmosphere. The amount of solar radiation that ultimately reaches Earth's surface limits the possible photosynthetic output of ecosystems.

However, only a small fraction of the sunlight that reaches Earth's surface is actually used in photosynthesis. Much of the radiation strikes materials that don't photosynthesize, such as ice and soil. Of the radiation that does reach photosynthetic organisms, only certain wavelengths are absorbed by photosynthetic pigments (see Figure 10.9); the rest is transmitted, reflected, or lost as heat. As a result, only about 1% of the visible light that strikes photosynthetic organisms is converted to chemical energy. Nevertheless, Earth's primary producers create about 150 billion metric tons 1.50×1014kg of organic material each year.

Gross and Net Production

Total primary production in an ecosystem is known as that ecosystem's **gross primary production** ** (GPP)**—the amount of energy from light (or chemicals, in chemoautotrophic systems) converted to the

chemical energy of organic molecules per unit time. Not all of this production is stored as organic material in the primary producers because they use some of the molecules as fuel for their own cellular respiration. **Net primary production (NPP)** is equal to gross primary production minus the energy used by the primary producers (autotrophs) for their cellular respiration (Ra, where "a" stands for autotrophs):

$$NPP = GPP - Ra$$

On average, NPP is about one-half of GPP. To ecologists, NPP is the key measurement because it represents the storage of chemical energy that will be available to consumers in the ecosystem. Using the analogy of a paycheck, you can think of net primary production (NPP) as the take-home pay, which equals gross primary production (GPP), the gross pay, minus respiration (Ra), the taxes.

Net primary production can be expressed as energy per unit area per unit time J/m2·yr or as biomass (mass of vegetation) added per unit area per unit time g/m2·yr. (Note that biomass is usually expressed in terms of the dry mass of organic material.) An ecosystem's NPP should not be confused with the total biomass of photosynthetic autotrophs present. The net primary production is the amount of *new* biomass added in a given period of time. Although the total biomass of a forest is large, its NPP may actually be less than that of some grasslands; grasslands do not accumulate as much biomass as forests because animals consume the plants rapidly and because grasses and herbs decompose more quickly than trees do.

Satellites provide a powerful tool for studying global patterns of primary production. Images produced from satellite data show that different ecosystems vary considerably in their NPP. For example, tropical rain forests are among the most productive terrestrial ecosystems and contribute a large portion of the planet's NPP **(Figure 55.5)**. Estuaries and coral reefs also have very high NPP, but their contribution to the global total is smaller because these ecosystems cover only about one-tenth the area covered by tropical rain forests. In contrast, while the open oceans are relatively unproductive, their vast size means that together they contribute as much global NPP as terrestrial systems do.

Figure 55.5

Global net primary production.

A map of satellite-collected data colorized to show net primary production. The highest rates of production are shown to be in the tropical areas such as the Amazon river basin.

The map is based on satellite-collected data, such as amount of sunlight absorbed by vegetation. Note that tropical land areas have the highest rates of production (yellow to red).

The color spectrum is correlated to net primary production as follows: 0 = black; 1 = blue; 2 = light green; 3 = red. Major desert regions are black, gray or purple; oceans are all purple; most land masses are blue (such as North America) however tropical and heavily forested areas are green, yellow and red.

VISUAL SKILLS *Does this map accurately reflect the significance of wetlands, coral reefs, and coastal zones, all highly productive habitats? Explain.*

*For suggested answer, see*Appendix A.**

Whereas NPP can be expressed as the amount of new biomass added by producers in a given period of time, **net ecosystem production (NEP)** is a measure of the biomass accumulation by producers *and* consumers during that time. NEP is defined as gross primary production minus the total respiration of all organisms in the system (RT)—not just primary producers, as for the calculation of NPP, but decomposers and other heterotrophs as well:

$$NEP = GPP - RT$$

NEP is useful to ecologists because its value determines whether an ecosystem is gaining or losing carbon over time. A forest may have a positive NPP but still lose carbon if heterotrophs release it as CO_2 more quickly than primary producers incorporate it into organic compounds.

The most common way to estimate NEP is to measure the net flux (flow) of CO_2 or O_2 entering or leaving the ecosystem. If more CO_2 enters than leaves, the system is storing carbon. Because O_2 release is directly coupled to photosynthesis and respiration (see Figure 9.1), a system that is giving off O_2 is also storing carbon. On land, ecologists typically measure only the net flux of CO_2 from ecosystems because detecting small changes in O_2 flux in a large atmospheric O_2 pool is difficult.

Next, we'll examine factors that limit production in ecosystems, focusing first on aquatic ecosystems.

Primary Production in Aquatic Ecosystems

In aquatic (marine and freshwater) ecosystems, both light and nutrients are important in controlling primary production.

Light Limitation

Because solar radiation drives photosynthesis, you would expect light to be a key variable in controlling primary production in oceans. Indeed, the depth of light penetration affects primary production throughout the photic zone of an ocean or lake (see Figure 52.14). About half of the solar radiation is absorbed in the first 15m of water. Even in "clear" water, only 5–10% of the radiation may reach a depth of 75m.

If light were the main variable limiting primary production in the ocean, you would expect production to increase along a gradient from the poles toward the equator, which receives the greatest intensity of light. However, you can see in Figure 55.5 that there is no such gradient. What other factor strongly influences primary production in the ocean?

Nutrient Limitation

More than light, nutrients limit primary production in most oceans and lakes. A **limiting nutrient** is the element that must be added for production to increase. The nutrients that most often limit marine production are nitrogen and phosphorus. Concentrations of these nutrients are typically low in the photic zone because they are rapidly taken up by phytoplankton and because detritus tends to sink.

In one study, detailed in **Figure 55.6**, nutrient enrichment experiments found that nitrogen was limiting phytoplankton growth off the south shore of Long Island, New York. Moreover, as also seen in Figure 55.6, primary production can increase dramatically when the nutrient status of an ecosystem changes from nutrient-poor to nutrient-rich, a process known as **eutrophication** (from the Greek *eutrophos*, well nourished). One practical application of this work is in preventing algal blooms caused by excess nitrogen runoff that fertilizes the phytoplankton. Preventing such blooms is critical because their occurrence can lead to the formation of large marine "dead zones," regions in which oxygen concentrations drop to levels that are fatal to many organisms (see Figure 56.24).

Figure 55.6

Inquiry: Which nutrient limits phytoplankton production along the coast of Long Island?

Experiment

Pollution from duck farms concentrated near Moriches Bay adds both nitrogen and phosphorus to the coastal water off Long Island, New York. To determine which nutrient limits phytoplankton growth in this area, John Ryther and William Dunstan, of the Woods Hole Oceanographic Institution, cultured the phytoplankton *Nannochloris atomus* with water collected from several sites, identified as A through G. They added either ammonium (NH_4^+) or phosphate (PO_4^{3-}) to some of the cultures.

Results

The addition of ammonium caused heavy phytoplankton growth in the cultures, but the addition of phosphate did not.

A bar graph plots phytoplankton density versus collection site. Each site shows three bars for ammonium enriched, phosphate enriched, and unenriched control. At all sites, ammonium resulted in heavy growth where the others did not.

Conclusion

Nitrogen is the nutrient that limits phytoplankton growth in this ecosystem because adding phosphorus did not increase *Nannochloris* growth, whereas adding nitrogen increased phytoplankton density dramatically.

Data from J. H. Ryther and W. M. Dunstan, Nitrogen, phosphorus, and eutrophication in the coastal marine environment, *Science*171:1008–1013 (1971).

WHAT IF? *Predict how the results would change if water samples were drawn from areas where new duck farms had greatly increased the amount of pollution in the water. Explain.*

*For suggested answer, see*Appendix A.**

The macronutrients nitrogen and phosphorus are not the only nutrients that limit aquatic production. Several large areas of the ocean have low phytoplankton densities despite relatively high nitrogen concentrations. The Sargasso Sea, a subtropical region of the Atlantic Ocean, has some of the clearest water in the world because of its low phytoplankton density. Nutrient enrichment experiments have revealed that the availability of the micronutrient iron limits primary production there **(Table 55.1)**. Windblown dust from land supplies most of the iron to the oceans but is relatively scarce in the Sargasso Sea and certain other regions compared to the oceans as a whole.

Table 55.1

Nutrient Enrichment Experiment for Sargasso Sea

A table titled Table 55.1 Nutrient Enrichment Experiment for Sargasso Sea illustrates the relative uptake of 14 C by cultures for nutrients added to the experimental culture.

Nutrients Added to Experimental Culture	Relative uptake of superscript 14 C by cultures superscript asterisk
None(controls)	1.00
Nitrogen (N) plus phosphorus (P) only	1.10
N plus P plus metals, excluding iron (F e)	1.08
N plus P plus metals, including F e	12.90
N plus P plus F e	12.00

*14C uptake by cultures measures primary production.

Data from D. W. Menzel and J. H. Ryther, Nutrients limiting the production of phytoplankton in the Sargasso Sea, with special reference to iron, *Deep Sea Research* 7:276–281 (1961).

INTERPRET THE DATA *The element molybdenum*(Mo)*is another micronutrient that can limit primary production in the oceans. If the researchers found the following results for additions of*Mo*, what would you conclude about its relative importance for growth?* N+P+Mo:6.0N+P+Fe+Mo:72.0 **

On the flip side, areas of *upwelling*, where deep, nutrient-rich waters circulate to the ocean surface, have exceptionally high primary production. This fact supports the hypothesis that nutrient availability determines marine primary production. Because upwelling stimulates growth of the phytoplankton that form the base of marine food webs, upwelling areas typically host highly productive, diverse ecosystems and are prime fishing locations. The largest areas of upwelling occur in the Southern Ocean (also called the Antarctic Ocean), along the equator, and in the coastal waters off Peru, California, and parts of western Africa.

Nutrient limitation is also common in freshwater lakes. During the 1970s, scientists showed that the sewage and fertilizer runoff from farms and lawns adds considerable nutrients to lakes, promoting the growth of primary producers. When the primary producers die, their bodies are broken down by aerobic decomposers. This depletes the water of much or all of its oxygen, killing large numbers of fish and causing the loss of many fish species from the lakes.

To prevent such fish kills, scientists need to know which nutrient is responsible. While nitrogen rarely limits primary production in lakes, many whole-lake experiments showed that phosphorus availability limited cyanobacterial growth. This and other ecological research led to the use of phosphate-free detergents and other water quality reforms.

Primary Production in Terrestrial Ecosystems

At regional and global scales, temperature and moisture are the main factors controlling primary production in terrestrial ecosystems. Primary production is greater in wetter ecosystems, as shown for the plot of NPP and annual precipitation in **Figure 55.7**. NPP also increases with temperature and the amount of solar energy available to drive evaporation and transpiration. As a result, tropical rain forests, with their warm, wet conditions that promote plant growth, are the most productive terrestrial ecosystems. In contrast, low-productivity systems are generally hot and dry, like many deserts, or cold and dry, like arctic tundra. Between these extremes lie the temperate forest and grassland ecosystems, with moderate climates and intermediate productivity.

Figure 55.7

A global relationship between NPP and mean annual precipitation for terrestrial ecosystems.

A graph plots net annual primary production versus mean annual precipitation. A scatterplot is modeled by a smooth curve that rises from (0, 0) to (200, 1175) with decreasing steepness. All values are estimated.

Nutrients can also affect NPP in terrestrial ecosystems. A dramatic example occurred after the arctic fox (see Figure 55.1) was introduced onto islands near Alaska. The presence of foxes had the surprising effect of converting grasslands to tundra, thereby lowering NPP. The foxes fed voraciously on the islands' seabirds, decreasing their density almost 100-fold. Fewer seabirds meant less bird guano (waste), a primary source of essential nutrients for plants on the islands. Researchers suspected that the scarcity of nutrients reduced the growth of nutrient-hungry grasses, favoring instead the slower-growing plants typical of tundra. To test this explanation, they added fertilizer to plots of tundra on one of the fox-infested islands. Three years later, the fertilized plots had reverted back to grassland.

Nutrient Limitations and Adaptations That Reduce Them

EVOLUTION As in aquatic systems, nitrogen and phosphorus are the nutrients that most commonly limit terrestrial production. Globally, nitrogen limits plant growth most. Phosphorus limitations are common in older soils where phosphate molecules have been leached away by water, such as in many tropical ecosystems. Note that adding a nonlimiting nutrient, even one that is scarce, will not stimulate production. Conversely, adding more of the limiting nutrient will increase production until some other nutrient becomes limiting.

Various adaptations have evolved in plants that can increase their uptake of limiting nutrients. One important adaptation is the mutualism between plant roots and nitrogen-fixing bacteria. Another is the mycorrhizal association between plant roots and fungi that supply phosphorus and other limiting elements to plants (see Figure 37.14). Plant roots also have hairs and other anatomical features that increase the area of soil in contact with the roots (see Figures 33.8 and 35.3). Many plants release enzymes and other substances into the soil that increase the availability of limiting nutrients; such substances include phosphatases, which cleave a phosphate group from larger molecules, and certain molecules (called chelating agents) that make micronutrients such as iron more soluble in the soil.

Effects of Climate Change on Production

As we've seen, climatic factors such as temperature and precipitation affect terrestrial NPP. Thus, we might expect that climate change could affect production in terrestrial ecosystems—and it does. For example, satellite data showed that from 1982 to 1999, NPP increased by 6% in terrestrial ecosystems. Nearly half of

this increase occurred in the tropical forests of the Amazon, where changing climate patterns had caused cloud cover to decrease, thereby increasing the amount of solar energy available to primary producers. Since 2000, however, these gains in NPP have been erased. This reversal was affected by another aspect of climate change: a series of major droughts in the Southern Hemisphere.

Effects of climate change on NPP can also be seen in the impact of "hotter droughts" on wildfires and insect outbreaks. Consider forests in the American Southwest. In recent decades, the forests of this region have experienced droughts driven by climate warming and changing patterns of precipitation. These ongoing droughts, in turn, have led to increases in the area burned by wildfires and the area affected by outbreaks of bark beetles such as the mountain pine beetle *Dendroctonus ponderosae***(Figure 55.8)**. As a result, tree mortality has increased and NPP has decreased in these forests.

Figure 55.8

Climate change, wildfires, and insect outbreaks.

Two graphs. Top graph shows area of trees burned by wildfire; bottom graph shows area of trees affected by bark beetles. Overall increase in drought stress index from 1997 to 2014.

Forests in the American southwest are experiencing hotter droughts caused by rising temperatures in the summer and reduced snowfall in the winter. The drought-stress index indicates how greatly trees are stressed by these conditions; rising values of this index correspond to increasing drought. Higher drought stress correlates with increasing area burned by wildfires (top) and affected by bark beetles (bottom), which target drought-stressed trees with weakened defenses.

Top graph: x-axis is labeled year and marked in the following increments: 1997, 2000, 2005, 2010, 2014. Y-axis is labeled area burned by wildfires (square kilometers, log scale) and marked as follows: 1, 10 squared, 10 cubed, 10 raised to the fourth power. The vertical axis on the right is labeled drought-stress index and marked as follows: negative 2, zero, plus 2. The area burned by wildfires is a blue line; the drought stress index is a red line. See table below for data for figure 55.8 top graph.

Bottom graph: x-axis is labeled year and marked in the following increments: 1997, 2000, 2005, 2010, 2014. Y-axis is labeled area affected by bark beetles (square kilometers, log scale) and marked as follows: 10 squared, 10 cubed, 10 raised to the fourth power. The vertical axis on the right is labeled drought-stress index and marked as follows: negative 1, zero, plus 1. The area affected by bark beetles is a blue line; the drought stress index is a red line. See table below for data for figure 55.8 bottom graph. For both graphs, the drought stress index largely parallels the area burned by wildfires and the area affected by bark beetles.

Climate change can also affect whether an ecosystem stores or loses carbon over time. As discussed earlier, net ecosystem production, or NEP, reflects the total biomass accumulation that occurs during a given period of time. When NEP>0, the ecosystem gains more carbon than it loses; such ecosystems store carbon and are

said to be a carbon *sink*. In contrast, when NEP<0, the ecosystem loses more carbon than it gains; such ecosystems are a carbon *source*.

Recent research shows that climate change can cause an ecosystem to switch from a carbon sink to a carbon source. For example, in some arctic ecosystems, climate warming has increased the metabolic activities of soil microorganisms, causing an uptick in the amount of CO_2 produced in cellular respiration. In these ecosystems, the total amount of CO_2 produced in cellular respiration now exceeds what is absorbed in photosynthesis. As a result, these ecosystems—which once were carbon sinks—are now carbon sources. When this happens, an ecosystem may contribute to climate change by releasing more CO_2 than it absorbs. Indeed, a 2017 study found that with climate warming, large regions of tundra in Alaska now release more CO_2 than they absorb—and for the years 2013 and 2014, the entire state of Alaska released more CO_2 than it absorbed. In the **Problem-Solving Exercise**, you can examine how outbreaks of an insect population may affect the NEP of forest ecosystems.

Problem-Solving Exercise

AP

Can an insect outbreak threaten a forest's ability to absorb CO_2 from the atmosphere?

One way to combat climate change is to plant trees, since trees absorb large amounts of CO_2 from the atmosphere, converting it to biomass through photosynthesis. But what happens to the carbon stored as biomass in trees when an insect population explodes in number? Such insect outbreaks have become more frequent with climate change.

Can an insect outbreak threaten a forest's ability to absorb C O 2 from the atmosphere?

One way to combat climate change is to plant trees, since trees absorb large amounts of C O 2 from the atmosphere, converting it to biomass through photosynthesis. But what happens to the carbon stored as biomass in trees when an insect population explodes in number? Such insect outbreaks have become more frequent with climate change. A tree is shown with dozens of pitch tubes, indications of a damaging outbreak of mountain pine beetles.

A tree with dozens of "pitch tubes," indications of a damaging outbreak of mountain pine beetles (inset).

In this exercise, you will test whether an outbreak of the mountain pine beetle (*Dendroctonus ponderosae*) alters the amount of CO_2 that a forest ecosystem absorbs from and releases to the atmosphere.

Your Approach The principle guiding your investigation is that every ecosystem both absorbs and releases CO_2. Net ecosystem production (NEP) indicates whether an ecosystem is a carbon sink (absorbing more CO_2 from the atmosphere than it releases; this occurs when NEP>0) or a carbon source (releasing more CO_2 than it absorbs; NEP<0). To find out if the mountain pine beetle affects NEP, you will determine a forest's NEP before and after a recent outbreak of this insect.

Your Data From 2000 to 2006, an outbreak of the mountain pine beetle killed millions of trees in British Columbia, Canada. The impact of such outbreaks on whether forests gain carbon (NEP>0) or lose carbon (NEP<0) was poorly understood. To find out, ecologists estimated net primary production (NPP) and cellular respiration by decomposers and other heterotrophs (Rh), before and after the outbreak. These data allow forest NEP to be calculated from the equation NEP=NPP−Rh.

A table illustrates before and after outbreak values of N P P, R subscript h, and N E P.

	N P P [g per (meter squared multiplied by year)]	**R subscript h [g per (meter squared multiplied by year)]**	**N E P [g per (meter squared multiplied by year)]**
Before outbreak	440	408	
After outbreak	400	424	

Your Analysis

- Complete the table by calculating the values of NEP before and after the outbreak. Before the outbreak, was the forest a carbon sink or a carbon source? After the outbreak?
- NEP is often defined as NEP=GPP−RT, where GPP is gross primary production and RT equals cellular respiration by autotrophs Ra*plus* cellular respiration by heterotrophs (Rh). Use the relation NPP=GPP−Ra to show that the two equations for NEP introduced in this exercise are equivalent.
- Based on your results in question 1, predict whether the mountain pine beetle outbreak could have feedback effects on the global climate. Explain.

Instructors: A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**.